MIL-S-60113(MU)
30 April 1964

MILITARY SPECIFICATION

SIGHTS, INFINITY REFLEX, FOR HELICOPTERS

1. SCOPE

1.1 This specification covers similar infinity reflex sights used on helicopters. These sights provide a reference for estimating and introducing lead angles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45208	Inspection System Requirements
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1270

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army Munitions Command

F8566264	Sights, Infinity Reflex: 8566264 and 8566264-1
F8566351	Sight, Infinity Reflex: 8566351
IEL 8566264	List of Inspection Documents for Sights Infinity Reflex: 8566264 and 8566264-1
IEL 8566351	List of Inspection Documents for Sight Infinity Reflex: 8566351

PACKAGING DATA SHEET

MIL-P-14232/P8566264	Packaging of Sight, Infinity Reflex: 8566264
MIL-P-14232/P8566264-1	Packaging of Sight, Infinity Reflex: 8566264-1
MIL-P-14232/P8566351	Packaging of Sight, Infinity Reflex: 8566351

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The sights shall be manufactured in accordance with the applicable drawings specified in Section 2 and drawings pertaining thereto.

3.2 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specifications MIL-O-13830 and MIL-F-13926:

3.2.1 Specification MIL-O-13830.

(a) Cleanliness.
(b) Lenses.

3.2.2 Specification MIL-F-13926.

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Electrical and electronic assemblies.
(e) Part identification and marking.
(f) Workmanship.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

3.3 Orientation.

3.3.1 Mount assembly.- The mounting face of the mount assembly shall be positioned in a vertical plane. The two holes for 5/16 inch mounting screws farthest from the elevation boresight adjustment screw shall be in the top position for Sights 8566264 and 8566264-1, and shall be in the bottom position for Sight 8566351.

3.3.1.1 Cross level adjustment.- The cross level adjustment shall be at mid-point of total travel within 1/32 inch.

3.3.1.2 Elevation adjustment.- The elevation boresight adjustment screw shall be positioned at mid-point of total adjustment range within one-half turn.

3.3.1.3 Azimuth adjustment.- The clearance at the spring-loaded azimuth adjustments shall not exceed .20 inch.

3.3.2 Adjustable reflector assembly.

3.3.2.1 Elevation knob.- The elevation knob shall be in detent.

3.3.2.2 Body.- The flat surface of Sight Housing 8643644, adjacent to Retainer 8643633, shall be horizontal within 1.0 mil in the axis perpendicular to the line of sight.

3.3.2.3 Inclinometer.- The inclinometer ball shall be centered within the width of one graduation line.

3.3.2.4 Elevation level vial.- The bubble in the elevation level vial shall be centered within the width of one graduation line.

3.4 Pre-vibration collimation.- Prior to meeting the vibration requirement of 3.5.4, the sight shall be positioned in accordance with 3.3. The line of sight, as defined by the center pip of the projected reticle pattern, shall be horizontal within 1.0 mil and perpendicular to the vertical mounting surface of the mounting bracket within 1.0 mil.

3.5 Environmental.

3.5.1 Storage temperature.- The sight shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90°F after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

3.5.2 Operating temperatures.- The sight shall meet the applicable requirements of 3.8 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

3.5.3 Sealing.- The sight shall show no evidence of leakage when subjected to an internal pressure of dry nitrogen of 5 pounds per square inch for a period of 5 minutes.

3.5.4 Vibration.- The sight shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 0.025 inch (0.05 inch total excursion) for a period of 10 ± 1 minutes. The lamp shall be energized and there shall be no evidence of flicker during vibration. Subsequent to vibration, there shall be no evidence of physical failure.

3.6 Post-environmental.- Subsequent to meeting the environmental requirements in 3.5 the sight shall meet the requirements of 3.7.1 to 3.11 inclusive. The requirements of 3.7.1 to 3.7.5 inclusive shall be met with the sight positioned in accordance with 3.3.

3.7 Accuracy.

3.7.1 Collimation.- The line of sight shall be within 1 mil in elevation and 0.5 mil in azimuth of the readings recorded in accordance with 3.4.

3.7.2 Reticle orientation.- The horizontal line of the projected reticle pattern shall be parallel to a horizontal target within 30 minutes.

3.7.3 Plumb travel.- The line of sight shall track a plumb line, through the excursion range specified in 3.8 2, within 1.0 mil. The maximum spread in aximuth shall not exceed 1.5 mils.

3.7.4 Elevation reading.- The error in elevation readings through the range specified in 3.8.2 shall not exceed 1.0°.

3.7.5 Backlash.- Backlash at any elevation setting shall not exceed 3.0 minutes through an excursion range of 13° in depression to 13° in elevation.

3.8 Operability.

3.8.1 Zero detent setting.- With the elevation knob in detent, the zero graduation line on the elevation knob shall be in coincidence with the zero graduation line on the MILS scale within the width of one graduation line of the elevation knob.

3.8.2 Excursion range.- Rotation of the elevation knob shall elevate line of sight a minimum of 14° depression to a minimum of 14° elevation.

3.8.3 Torque.- The requirements of 3.8.3.1 and 3.8.3.2 shall be met while the sight is exposed to the temperature requirements of 3.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

3.8.3.1 Elevation knob.- The torque required to rotate the elevation knob shall not be less than 5 nor more than 15 pound-inches.

3.8.3.2 Adjustable index.- The adjustable index shall be operable.

3.9 Reticle.- The image of the projected reticle shall be clearly defined and there shall be no irregularities in the pattern.

3.10 Cleanliness.- There shall be no dirt within the sight which adversely affects the image quality. The sight shall meet the cleanliness and optical quality requirements of Specification MIL-O-13830.

3.11 Parallax.- With the eye 9 to 11 inches from the center of the reflector plate, parallax at the center of the reticle image shall not exceed 0.2 mil when viewing a target from 500 to 2500 meters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the Contract or Purchase Order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in IEL 8566264 and IEL 8566351. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.5%			
101.	Collimation (pre-vibration)	3.4	4.5.2.2
102.	Vibration	3.5.4	4.5.2.3
103.	Collimation (Post)	3.7.1	4.5.2.4
104.	Reticle orientation	3.7.2	4.5.2.5
105.	Plumb travel	3.7.3	4.5.2.6
106.	Elevation reading	3.7.4	4.5.2.7
107.	Backlash	3.7.5	4.5.2.8
108.	Zero setting (detent)	3.8.1	4.5.2.9
109.	Cleanliness	3.10	4.5.2.11
110.	Parallax	3.11	4.5.2.12
MINOR: AQL 4.0%			
201.	Range limit	3.8.2	4.5.2.7.1
202.	Irregularities in reticle pattern	3.9	4.5.2.10

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

4.3 Control Tests.

4.3.1 General.- One sight of the type under contract shall be selected at random from each 100 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
Fabrication	3.1	Visual - per dwgs.
General Specification	3.2	Visual-MIL-F-13926 & MIL-O-13830

4.3.2 Environmental.- Three control samples shall be selected from the same production quantities specified in 4.3.1 and they shall be subjected to the tests outlined below:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
Storage temperature	3.5.1	4.5.1.1
Torque (elevation knob)	3.8.3.1	4.5.1.2.1
Adjustable index	3.8.3.2	4.5.1.2.2
Sealing	3.5.3	4.5.1.3

4.3.3 Control test failure.- Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor furnished inspection equipment.

4.4.1.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents IEL 8566264 and IEL 8566351. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL-I-45208.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

4.4.1.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.1.3 In addition to those accuracies specified above, the conditions outlined under "test facilities" specified in Specification MIL-F-13926 shall apply to test equipment used in performing the vibration and temperature test.

4.4.1.4 Test target accuracy.- The target Fig. I provides three sets of intersecting lines (A & B, A & C, and A & D). Selecting the proper lines depends on the type of sight under procurement. Target line A provides the means of aligning the target in azimuth, and target lines C and D provides the means for placing the target at the proper elevation when the collimating telescope, Drawing C5549108 is positioned in the "V" grove of Adapter, D7566785 and secured to Cross Leveling Fixture, D6523553.

4.4.1.4.1 The vertical displacement (X_1 & X_2) between target lines B & C and B & D shall be within the limits outlined on set-up instructions supplied with the Cross Leveling Fixture, D6523553.

4.4.1.4.2 Lines E and F are provided to measure parallelism between the sight's horizontal reticle line and target line B. The angular position of the target lines shall represent the tolerance specified in 3.7.2.

4.4.1.4.3 A two mil square shall be provided about the intersecting point of target lines A & B and is divided into .2 mil increments to facilitate measuring collimation error and possible reticle movement.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Storage temperatures.- With the sights placed in the test chamber, the temperature shall be gradually reduced to -80°F until thermally stabilized. Avoiding thermal shock, the sight shall then be subject to +160°F and thermally stabilized. Upon return to standard ambient the sights shall be given a visual and tactile examination to insure compliance with 3.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

4.5.1.2 Operating temperatures (Torque).

4.5.1.2.1 Elevation knob.- Elevation knob torque shall be measured at the standard ambient temperature (60 to 90°F) and at each of the extreme temperatures specified in 3.5.2 to determine compliance with 3.8.3.1. The test shall be performed utilizing a standard torque wrench with appropriate adapter.

4.5.1.2.2 Adjustable index.- This examination shall be conducted concurrently with 4.5.1.2.1. A tactile examination shall be made to insure that the index is adjustable at the standard ambient temperature and at each of the extreme operating temperatures of 3.5.2 to determine compliance with 3.8.3.2.

4.5.1.3 Sealing.- The sights shall meet the requirement of 3.5.3 when tested in accordance with the sealing test outlined in the set-up and operating instructions on Drawing F7680673.

4.5.2 Accuracy and operability.

4.5.2.1 Orientation.- The test to determine accuracy and operability specified in 3.7, 3.8 and 3.9 shall be performed with the sight oriented in accordance with 3.3. The test target (Fig. 1) shall be utilized for measuring accuracy and operability requirements. The set up instruction outlined on Drawing, B8566784 specify proper orientation of Cross Leveling Fixture, D6523553 and Adapter D8566785 and also the aligning of the test target to Adapter, D8566785. With the sight properly secured to the oriented Cross Leveling Fixture, the inclinometer shall be positioned as specified in 3.3.2.3 and the elevation level vial shall be positioned within the limits of 3.3.2.4. To determine accuracy of level vial readings and proper orientation of the mounting bracket, an Adapter: D8566787 and Inspection Aid (Gunners Quadrant: M1A1) Drawing A8658940 shall be utilized. The Gunners Quadrant shall be set to indicate zero. It shall be placed against the locating pins of the Adapter. Place the Adapter on the surface specified in 3.3.2.2. The Quadrant level vial shall indicate level within the limits of 3.3.2.2 to insure that the mounting bracket is oriented in accordance with 3.3.1 and that the elevation level vial and inclinometer on the sight are properly adjusted.

4.5.2.2 Pre-vibration collimation.- The sight shall be inspected to determine compliance with 3.4 by being oriented and set up in accordance with 4.5.2.1. Prior to vibration, collimation shall be within the tolerance specified in 3.4. Observe where the center reticle pip falls with respect to the intersecting target lines A-B (Figure 1) when the elevation knob is set in the zero detent. The positions of the line of sight through the geometric axis shall be recorded for comparison with the collimation setting determined in 4.5.2.4 to determine compliance with 3.7.1.

4.5.2.3 Vibration.- The complete sight shall have met the requirement of 3.4 prior to being vibrated. With the complete sight and Adapter Drawing D8566785 secured in the position defined on Drawing D8566784 to the vibration fixture, it shall be vibrated in a vertical plane in accordance with the amplitude, cycles and time duration specified in 3.5.4. During vibration the lamp shall be energized and

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

a visual examination made to determine conformance to the flicker requirement in 3.5.4. Subsequent to vibration the sight shall show no evidence of physical failure and shall be subjected to the following examinations and tests.

4.5.2.4 Collimation (post vibration). This test shall be performed after orientation has been established in accordance with 4.5.2.1 and utilizing Cross Leveling Fixture, D6523553 with Adapter, D7566785. Reticle movement is measured after the completion of the vibration test by securing Adapter, D8566785 to the Cross Leveling Fixture as outlined in Drawing D8566784, without disturbing any of the previous settings. Observe where the reticle center pip falls upon the target with respect to the recorded position. This second position must conform to the recorded position within the limit specified in 3.7.1.

4.5.2.5 Reticle orientation.- With the mount and sight positioned as outlined in 4.5.2.1 observe where the reticle horizontal line falls with respect to target line "B" (Figure 1). They shall be parallel to each other within the limits of target lines E & F to conform to the limits specified in 3.7.2.

4.5.2.6 Plumb travel.- With the sight and mount positioned as outlined in 4.5.2.1 plumb travel shall be determined by rotating the elevation knob in one direction and the elevation wheel of the cross Leveling Fixture an equal amount in an opposite direction. During this rotation, sight through the reflector and observe coincidence between the center pip of the reticle and target line A. This coincidence shall not exceed the limits of 3.7.3 when the center pip is moved through the full travel of the elevation knob.

4.5.2.7 Elevation reading.- With the sight and mount positioned as outlined in 4.5.2.1, elevation readings of 14 degrees elevation and 14 degrees depression shall be measured by rotating the elevation knob 14 degrees from the detent position as measured by the elevation scale. Rotate the elevation handwheel of the fixture in an opposite direction until the level vials indicate level. Read the fixtures elevation vernier and scale. This reading and the reading on the elevation knob shall coincide within the limit of 3.7.4. Repeat this procedure in the opposite direction.

4.5.2.7.1 Excursion range.- This test shall be conducted concurrently with 4.5.2 to determine compliance with 3.8.2.

4.5.2.8 Backlash.- With the mount and sight positioned as outlined in 4.5.2.1, backlash shall be tested by rotating the elevation handwheel of the fixture to any selected setting and rotating the sight elevation knob in an opposite direction until the reticle center pip coincides with the original zero position. Record this reading. Continue to rotate the sight elevation knob beyond this point and return the line of sight to the previous position without overtravel and observe the elevation reading on the elevation knob. This reading must coincide with the recorded reading within the limits of 3.7.5. Repeat this procedure at five other settings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

4.5.2.9 Zero detent setting.- Set the elevation knob to engage the detent and observe coincidence between the zero graduation lines on the elevation knob. They must coincide within the limits outlined in 3.8.1.

4.5.2.10 Reticle.- This test shall be performed by observing the reticle pattern while the sight is directed toward the targets specified above. The pattern shall conform to the requirement outlined in 3.9. This test shall be performed with the peepsight (specified on Dwg. 8566784) removed.

4.5.2.11 Cleanliness.- Inspection shall be conducted in accordance with appropriate test methods outlined in Specification MIL-O-13830.

4.5.2.12 Parallax.- This test can be accomplished by directing the line of sight toward a target point at the distance specified in 3.11. With the eye placed at the eye distance specified in 3.11 the observed parallax shall not exceed the limit specified in 3.11.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and the applicable Packaging Data Sheet specified in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The infinity reflex sights are primarily intended for use as gunsight or aiming devices on helicopters. The sights are a part of the following systems:

Sight 8566264	XM3 Armament Subsystem
Sight 8566264-1	XM3 Armament Subsystem
Sight 8566351	XM3 and XM22 Armament Subsystem

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of an applicable level of preservation, packaging, and packing.
(c) Applicable stock number.

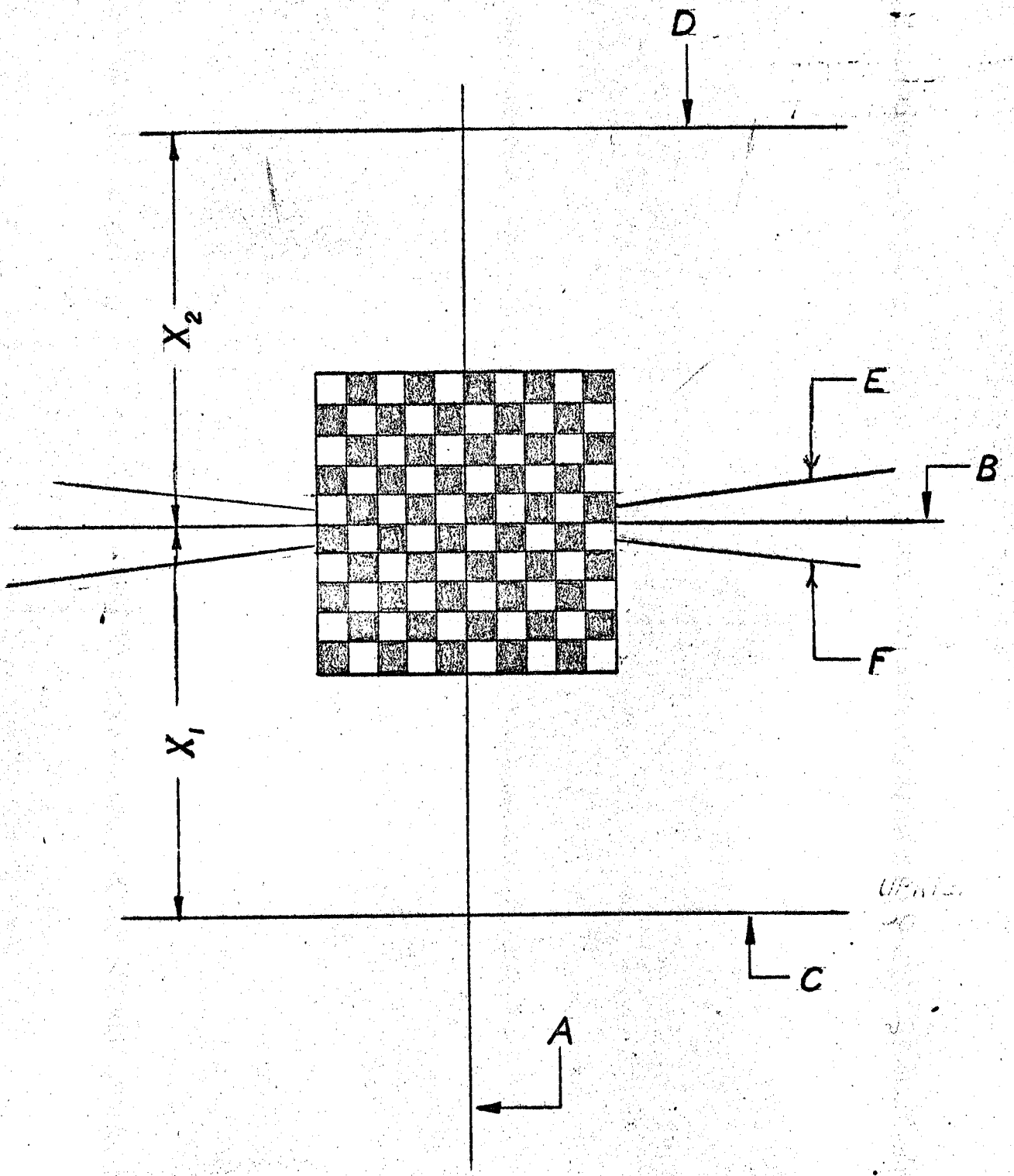
Custodian:

Army - MU

Preparing activity:

Army - MU

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.



TEST TARGET
FIGURE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.